CALEB BELLSMITH

BIOMEDICAL ENGINEERING STUDENT

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EDUCATION

University of Waterloo / Bachelor of Applied Science, Biomedical Engineering Co-op **Sept 2025 - Present**

- **WEEF Representative:** Nominated by class to evaluate technical proposals for student project funding and represent class interests.
- **Awards:** Governor General's Academic Medal, Sir Isaac Newton Physics Award, TDCH Valedictorian, HHCS Valedictorian.

TECHNICAL SKILLS

CAD/EDA: SolidWorks, Fusion360, AutoCAD, KiCad, Altium
Fabrication: CNC Machining, 3D Printing, Soldering, Manual Prototyping, Woodworking, Welding, Waterjetting, Laser Cutting
Programming: C++, C, Python, HTML, LaTeX, Visual Basic, MATLAB, CSS, JavaScript
Dev Tools: VSCode, CLion, Arduino IDE, Bamboo, GitHub, GitLab

EXPERIENCE

Automation Designer / Alchemy Nano **Sept 2026 - Present**
Part-Time

- Built a custom in-line scanning system for nanocomposite film quality testing, using an array of tiled global shutter cameras and a custom roll to roll converter.
- Created the full vision software that performs camera calibration and defect detection, guides focus and framing, and maps flagged regions into automated reports.
- Took the system through **complete production implementation**, including electrical, mechanical, and safety certifications.
- Optimized previously built automated micro graphing systems for production grade use.

Undergraduate Research Assistant / Wireless Sensors and Devices Laboratory **Sept 2026 - Present**
University of Waterloo / Research

- Developed autonomous assembly line robotics learning systems through mechanical optimization and custom PCB design for industrial automation settings.

Lab Automation Engineering Co-op / Alchemy Nano **Apr 2026 - Sept 2026**
Co-op

- Increased lab testing efficiency by up to **440%** and accuracy by **40%** by building an automated micro-graphing robot with a frame-quality classifier (**MobileNetV3**, trained on **1000+** images, **99.3%** consistency).
- Replaced manual optical measurement with Python CV tools for distortion and contact angle analysis utilizing gradient edge detection, parallel batch processing and **RANSAC** baseline fitting.
- Designed a motorized film take-up roller with quadrature encoder closed loop feedback, a touchscreen interface, and flash stored resumable cycles, built to Canadian electrical and workplace regulations.

Mechanical Designer / UWaterloo Space Soldering Team **2025 - Present**
Student Design Team

- Constructed a centrifuge for a microgravity soldering experiment selected for the **CAN-RGX Flight Campaign 2026** (NRC & CSA partnership); the technology is **patent pending** with commercial applications in space manufacturing and repair.
- Designed and fabricated the wire feeding assembly, slip ring mounting structure, and core rotation mechanism, validated through **FEA** and preliminary integration tests on all systems.

Electrical Designer / True North Bio-competition **2025 - 2026**
Student Design Team

- Designed the **custom PCB** serving as the complete electrical control framework for a lower limb exoskeleton, routing 3-channel EMG into a Teensy for high-speed ADC and an ESP32 for BLE transmission, with USB-C power filtering and noise-suppression decoupling.

PROJECTS

Jolt Mouse (Hack Canada 2025)

- Created a wearable human machine interface for amputee and upper limb disabled users, using IMU tilt for cursor control and EMG flexion for clicks over BLE, with web dashboard to monitor and tune sensitivity and EMG positioning.

Precision Robotic Arm

- Designed and hand built a **5 DoF** robotic arm with a **1 N-m** grip, USB-C PD charging from a laptop charger, and calibrated joystick proportional control, now being integrated with an EMG sleeve for motion mirroring.

Tilting Maze Platform

- Built a dual-axis servo platform commanded by a Python CV pipeline (HSV ball tracking, Artificial Potential Field navigation, **adaptive PID** auto-tuned every 60 frames, EMA velocity predictor) with on-disk trajectory learning to optimize solve time.

Custom Cedar Strip Canoe

- Computer modeled and fabricated a custom canoe using CNC machined internal molds and paired with traditional woodworking.